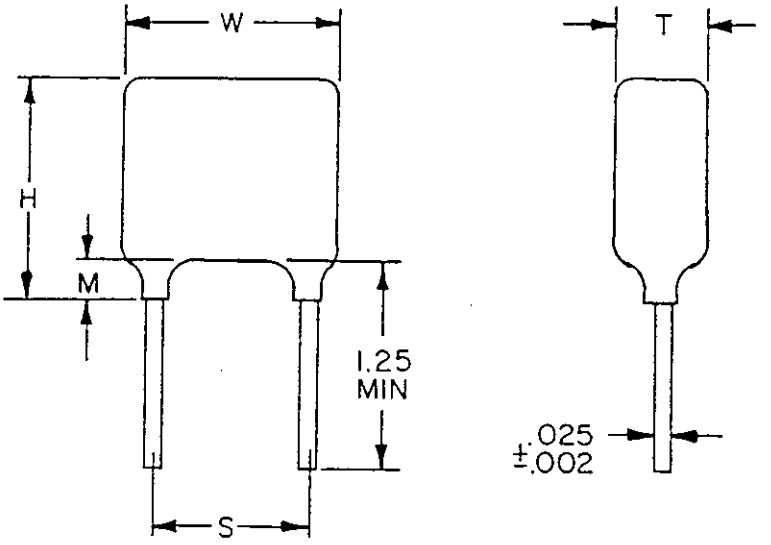
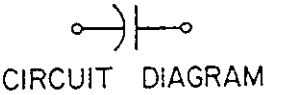
MILITARY SPECIFICATION SHEET

# Capacitors, Ceramic, Multilayer, High Voltage, 3000 V DC, Established Reliability, Style HVR03

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49467.




| Case code | Dimensions | | | | |
|---|---|---|---|---|---|
| | W max | H max | T max | M max | S ±.030 |
| B | .320 | .280 | .250 | .045 | .220 |
| C | .370 | .300 | .250 | .045 | .275 |
| D | .470 | .400 | .270 | .060 | .375 |
| E | .570 | .500 | .270 | .060 | .475 |
| F | .670 | .600 | .270 | .060 | .575 |
| G | .770 | .720 | .270 | .060 | .675 |
| J | 1.300 | .600 | .270 | .060 | 1.175 |
| K | 1.500 | .720 | .270 | .060 | 1.375 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

| Inches | mm | Inches | mm |
|---|---|---|---|
| .002 | .05 | .470 | 11.94 |
| .025 | .64 | .475 | 12.06 |
| .030 | .76 | .500 | 12.70 |
| .045 | 1.14 | .570 | 14.48 |
| .060 | 1.52 | .575 | 14.60 |
| .220 | 5.59 | .600 | 15.24 |
| .250 | 6.35 | .670 | 17.02 |
| .270 | 6.86 | .675 | 17.14 |
| .275 | 6.98 | .720 | 18.29 |
| .280 | 7.11 | .770 | 19.56 |
| .300 | 7.62 | 1.175 | 29.84 |
| .320 | 8.13 | 1.250 | 31.75 |
| .370 | 9.40 | 1.300 | 33.02 |
| .375 | 9.52 | 1.375 | 34.92 |
| .400 | 10.16 | 1.500 | 38.10 |

FIGURE 1. Capacitor style HVR03.


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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Conformally coated, molded, or encapsulated.

Lead material: In accordance with MIL-STD-1276.

Capacitance value: See table I.

Capacitance tolerance: J = ±5%, K = ±10%, M = ±20%. See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: 3000 V dc.

Failure rate level: M (1.0%), P (0.1%), R (.01%), or S (.001%).

Dielectric types: BP, BR, and BZ in accordance with MIL-C-49467.

Temperature coefficient: 0 ±30 ppm/°C (BP), ±15% (BR, BZ).

Voltage-temperature limits: In accordance with MIL-C-49467.

| Symbol | Capacitance change with reference to +25°C | |
|---|---|---|
| | Steps A through D of voltage-temperature limit cycle table of MIL-C-49467. Bias = 0 volt. | Steps E through G of voltage-temperature limit cycle table of MIL-C-49467. |
| BP | 0 ±30 ppm/°C | 100% rated voltage: 0 ±30 ppm/°C |
| BR | ±15% | 100% rated voltage: +15, -40%<br>40% rated voltage: +15, -25%<br>60% rated voltage: +15, -40% |
| BZ | ±15% | |

Dissipation factor (25°C): 2.5% maximum (BR and BZ characteristics) or 0.1% maximum (BP characteristic), measured under the same conditions as capacitance.

Resistance to soldering heat: $\Delta C$: -1% to +2% for BP characteristic.
-1% to +6% for BR and BZ characteristics.

Marking: In accordance with MIL-C-49467.

APPLICATION NOTES:

Additional encapsulation is necessary in applications where the possibility of a voltage breakdown between the leads of the capacitor, or the capacitor to another potential, could occur.

Heat sinks on each lead or adequate preheating is required when these capacitors are installed in or removed from circuits by soldering iron.



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TABLE I. Capacitor characteristics.

| PIN 1/ | Capacitance | Capacitance tolerance | Characteristic | Case code |
|---|---|---|---|---|
| M49467P03100__ | 10 pF | J, K | BP | B |
| M49467P03120__ | 12 | J, K | BP | B |
| M49467P03150__ | 15 | J, K | BP | B |
| M49467P03180__ | 18 | J, K | BP | B |
| M49467P03220__ | 22 | J, K | BP | B |
| M49467P03270__ | 27 | J, K | BP | B |
| M49467P03330__ | 33 | J, K | BP | B |
| M49467P03390__ | 39 | J, K | BP | B |
| M49467P03470__ | 47 | J, K | BP | B |
| M49467P03560__ | 56 | J, K | BP | B |
| M49467P03680__ | 68 | J, K | BP | B |
| M49467P03820__ | 82 | J, K | BP | B |
| M49467P03101__ | 100 | J, K | BP | B |
| M49467_03101__ | 100 | K, M | BR, BZ | B |
| M49467P03121__ | 120 | J, K | BP | B |
| M49467_03121__ | 120 | K, M | BR, BZ | B |
| M49467P03151__ | 150 | J, K | BP | B |
| M49467_03151__ | 150 | K, M | BR, BZ | B |
| M49467P03181__ | 180 | J, K | BP | B |
| M49467_03181__ | 180 | K, M | BR, BZ | B |
| M49467P03221__ | 220 | J, K | BP | B |
| M49467_03221__ | 220 | K, M | BR, BZ | B |
| M49467P03271__ | 270 | J, K | BP | B |
| M49467_03271__ | 270 | K, M | BR, BZ | B |
| M49467P03331__ | 330 | J, K | BP | B |
| M49467_03331__ | 330 | K, M | BR, BZ | B |
| M49467P03391__ | 390 | J, K | BP | B |
| M49467_03391__ | 390 | K, M | BR, BZ | B |
| M49467P03471__ | 470 | J, K | BP | B |
| M49467_03471__ | 470 | K, M | BR, BZ | B |
| M49467P03561__ | 560 | J, K | BP | C |
| M49467_03561__ | 560 | K, M | BR, BZ | B |
| M49467P03681__ | 680 | J, K | BP | C |
| M49467_03681__ | 680 | K, M | BR, BZ | B |
| M49467P03821__ | 820 | J, K | BP | D |
| M49467_03821__ | 820 | K, M | BR, BZ | B |
| M49467P03102__ | 1000 | J, K | BP | D |
| M49467_03102__ | 1000 | K, M | BR, BZ | B |
| M49467P03122__ | 1200 | J, K | BP | D |
| M49467_03122__ | 1200 | K, M | BR, BZ | B |
| M49467P03152__ | 1500 | J, K | BP | D |
| M49467_03152__ | 1500 | K, M | BR, BZ | B |
| M49467P03182__ | 1800 | J, K | BP | E |
| M49467R03182__ | 1800 | K, M | BR | C |
| M49467Z03182__ | 1800 | K, M | BZ | B |
| M49467P03222__ | 2200 | J, K | BP | E |
| M49467R03222__ | 2200 | K, M | BR | C |
| M49467Z03222__ | 2200 | K, M | BZ | B |
| M49467P03272__ | 2700 | J, K | BP | E |
| M49467R03272__ | 2700 | K, M | BR | D |
| M49467Z03272__ | 2700 | K, M | BZ | B |

See footnote at end of table.



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TABLE I. Capacitor characteristics - Continued.

| PIN 1/ | Capacitance | Capacitance tolerance | Characteristic | Case code |
|---|---|---|---|---|
| M49467P03332__ | 3300 pF | J, K | BP | E |
| M49467R03332__ | 3300 | K, M | BR | D |
| M49467Z03332__ | 3300 | K, M | BZ | C |
| M49467P03392__ | 3900 | J, K | BP | F |
| M49467R03392__ | 3900 | K, M | BR | D |
| M49467Z03392__ | 3900 | K, M | BZ | C |
| M49467P03472__ | 4700 | J, K | BP | F |
| M49467 03472__ | 4700 | K, M | BR, BZ | D |
| M49467P03562__ | 5600 | J, K | BP | G |
| M49467 03562__ | 5600 | K, M | BR, BZ | D |
| M49467P03682__ | 6800 | J, K | BP | G |
| M49467 03682__ | 6800 | K, M | BR, BZ | D |
| M49467R03822__ | 8200 | K, M | BR | E |
| M49467Z03822__ | 8200 | K, M | BZ | D |
| M49467R03103__ | .01 µF | K, M | BR | E |
| M49467Z03103__ | .01 | K, M | BZ | D |
| M49467R03123__ | .012 | K, M | BR | E |
| M49467Z03123__ | .012 | K, M | BZ | D |
| M49467R03153__ | .015 | K, M | BR | F |
| M49467Z03153__ | .015 | K, M | BZ | E |
| M49467R03183__ | .018 | K, M | BR | F |
| M49467Z03183__ | .018 | K, M | BZ | E |
| M49467R03223__ | .022 | K, M | BR | F |
| M49467Z03223__ | .022 | K, M | BZ | E |
| M49467R03273__ | .027 | K, M | BR | G |
| M49467Z03273__ | .027 | K, M | BZ | F |
| M49467R03333__ | .033 | K, M | BR | G |
| M49467Z03333__ | .033 | K, M | BZ | F |
| M49467 03393__ | .039 | K, M | BR, BZ | G |
| M49467R03473__ | .047 | K, M | BR | J |
| M49467Z03473__ | .047 | K, M | BZ | G |
| M49467R03563__ | .056 | K, M | BR | J |
| M49467Z03563__ | .056 | K, M | BZ | G |
| M49467R03683__ | .068 | K, M | BR | K |
| M49467Z03683__ | .068 | K, M | BZ | J |
| M49467 03823__ | .082 | K, M | BR, BZ | K |
| M49467 03104__ | .10 | K, M | BR, BZ | K |

1/ The complete PIN shall include additional letter symbols to indicate the characteristic, where applicable, the capacitance tolerance, and the failure rate level.



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## CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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